

26
18
44
13
57
2
65
7
73

COMP 315: Design and Analysis of Algorithms

CAT 1

Date: 14/10/24

Attempt all questions

1. Solve the job sequencing problem given $n = 6$, profits ^{18 27 34 5 2} (2, 7, 9, 26, 18, 13) and deadlines (1, 3, 2, 1, 5, 4) using greedy strategy. (5 marks)
2. State and explain the difference between dynamic programming and divide and conquer. (5 marks)
3. Discuss how posteriori analysis is carried out. (5 marks)
4. What is minimum cost spanning tree? Explain Prim's algorithm as applied to solving minimum cost spanning tree problem. (5 marks)
5. State and explain the characteristics of a good algorithm. (5 marks)
6. Using a theorem, explain Big-Oh notation. (5 marks)